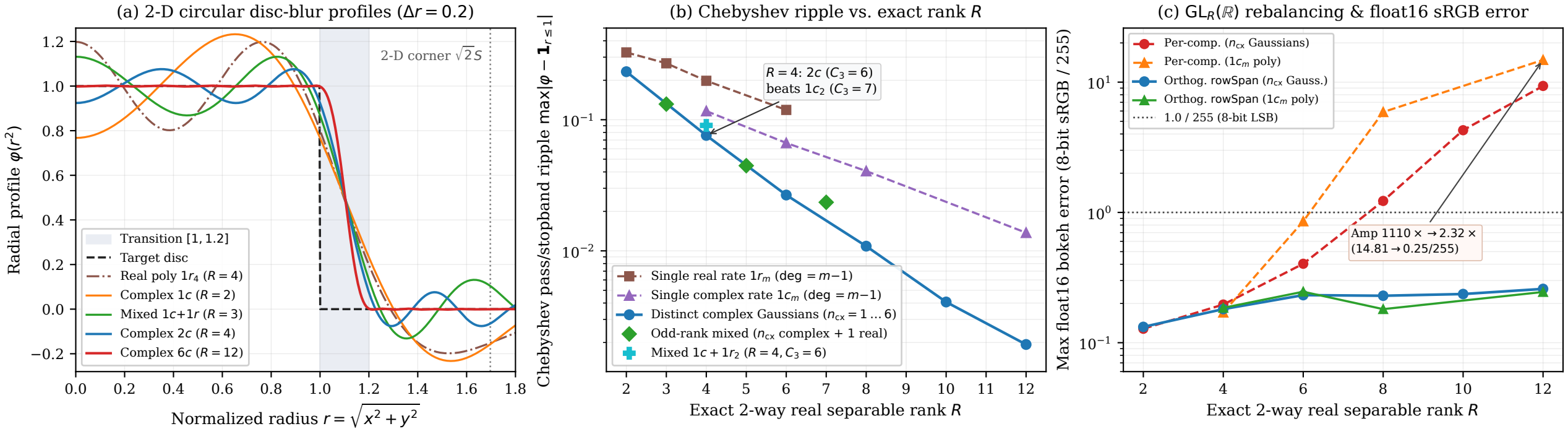

(a) 2-D circular disc-blur profiles ($\Delta r = 0.2$)
2-D corner $\sqrt{2}s$
Transition [1, 1.2]
Target disc
Real poly $1r_4$ ($R = 4$)
Complex $1c$ ($R = 2$)
Mixed $1c+1r$ ($R = 3$)
Complex $2c$ ($R = 4$)
Complex $6c$ ($R = 12$)
Radial profile $\varphi(r^2)$
Normalized radius $r = \sqrt{x^2 + y^2}$
(b) Chebyshev ripple vs. exact rank R
$R = 4$: $2c$ ($C_3 = 6$)
beats $1c_2$ ($C_3 = 7$)
Single real rate $1r_m$ (deg $= m-1$)
Single complex rate $1c_m$ (deg $= m-1$)
Distinct complex Gaussians ($n_{cx} = 1 \dots 6$)
Odd-rank mixed (n_{cx} complex + 1 real)
Mixed $1c + 1r_2$ ($R = 4, C_3 = 6$)
Chebyshev pass/stopband ripple $\max|\varphi - \mathbf{1}_{r \le 1}|$
Exact 2-way real separable rank R
(c) $\mathrm{GL}_R(\mathbb{R})$ rebalancing & float16 sRGB error
Per-comp. (n_{cx} Gaussians)
Per-comp. ($1c_m$ poly)
Orthog. rowSpan (n_{cx} Gauss.)
Orthog. rowSpan ($1c_m$ poly)
1.0 / 255 (8-bit LSB)
Amp 1110 × → 2.32 ×
(14.81 → 0.25/255)
Max float16 bokeh error (8-bit sRGB / 255)
Exact 2-way real separable rank R